

Inspection of thinking as a source of ideas for understanding minds

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Understanding minds

- It'd be cool to understand minds better.
- The field of AI cares about this given that its central question is:
 - How can one make artificial minds, ie intelligent beings, ie systems that can do a lot of different tasks (like humans), ie agents?
- Understanding minds is also important for AI alignment because the field concerns itself with making certain kinds of minds.
- More precisely, at least aspirationally, understanding minds could help us do the following:
 - design some sort of well-understood AI whose effects can be precisely pre-specified
 - ditto but for an AI with various opaque components which nevertheless do specific understood things
 - understand the thoughts that some AIs are thinking
 - push scaling lab AIs toward being nicer
 - control potentially misaligned AIs
 - understand whether such and such an AGI/ASI would be bad to humans
 - make us able to convince other researchers, elites, the public, and governments to ban AGI (if this is warranted)

A good subquestion: “how does (human) thinking work?”

- So, we are interested in understanding how impressive AIs would/could work.
- Fortunately, we have impressive systems in front of us: humans.
 - These days, we also have pretty impressive AIs in front of us.
- We can try to figure out how these systems do big things.
- These fields have an aspiration to answer this question:
 - philosophy, psychology, cognitive science, neuroscience, AI, interpretability, maybe economics

Sutskever on humans as a source of ideas in AI

- [Dwarkesh Patel to Ilya Sutskever](#): “What is research taste? ... What is it, how do you characterize how you come up with these ideas?”
- Ilya Sutskever: “I can comment on this for myself. I think different people do it differently. One thing that guides me personally is an aesthetic of how AI should be, by thinking about how people are.”
- There are two different important things here:
 - You can look at human brains.
 - eg Sutskever mentions the example of neurons and connections in brains
 - Or you can look at human cognition at a higher level.
 - eg Sutskever speaks about how as a human learning to drive, you have a robust sense of how well you are doing, and speculates this is like an RL value function, but of a sort we can't really write down yet

This talk/workshop is about looking at human cognition at a higher level.

Another good subquestion: “how should thinking work?”

- To understand thinking, one can also ask “should” questions like:
 - How could/should one solve this problem?
 - Can I write down pseudocode solving this class of problems?
 - What advice would I give someone to help them solve this class of problems? What would I teach them?
 - How could I have solved this problem faster?
- Human thinking provides one answer to “how should thinking work?”, given that it has been optimized to work well by various “forces”:
 - evolution has done a bunch, eg improving neuron connection “update rules” or creating some structures supporting language use
 - cultural evolution/accumulation has done a bunch, eg giving us all our words and the associated concepts, giving us tools, technologies, writing, various methods in science
 - lifetime learning has done a bunch, eg giving you various idiosyncratic good questions to ask, giving you various concepts and making you a capable user of human concepts

Historical examples of concepts and pieces of understanding about thinking gained in these ways

- belief
- proposition
- logic, implication, inference, argument, proof
- probability, random variable
- concept, definition
- meaning
- compositionality
- analogy, model, representation
- method, procedure, algorithm
- policy, observation, prediction, action
- planning, utility (in the decision-theoretic sense), reward, reinforcement learning, search, tree search in games
- optimization
- extremely many more specific versions of these things

I think there is probably a lot more to be discovered this way.

An investigation you should try

I recommend trying the following investigation:

- Try to inspect your own thinking, particularly when solving some problem. When doing this, keep in mind the goal of getting useful ideas for writing (pseudo)code of a pretty well understood problem-solver AI.

I recommend this for personal learning and for trying to push human understanding of minds forward (though this is of course nontrivial). (Obviously, this is not to say you should not also be doing other things, such as reading textbooks, other books, papers, posts relevant to how minds work.)

A case for this sort of investigation over interp, for understanding minds:

- Tsvi Benson-Tilsen, [Koan: divining alien datastructures from RAM activations](#)

The main exercise type for the workshop part today

- You will try inspecting your thinking when solving fairly short technical problems.
 - I will mention some other versions of inspecting thinking at the end of the talk.
- You should have this as a background goal: try to get ideas for writing pseudocode of sth like a GOFAL.
- The suggested procedure is:
 - As you're solving the problem, try to write down your chain of thought in basically as much detail as possible, at least for some parts. (Feel free to become more lazy on parts where you feel like you'd just write down the same thing.)
 - At the same time, try to speculate about how you're doing this thinking. Feel free to stop thinking about the object-level thing for some time to speculate further. Provide hypotheses and ask questions.
 - Once you're done, read your transcript again and see if you can identify more interesting things or questions.
 - Where did you make mistakes? How could you have solved the problem faster? What abstract advice would you like to have had for solving the problem better?

Fermi estimation inspection exercise

Run the inspection protocol on solving the following Fermi estimation problem:

- How much energy did training GPT-4 take? (You can give the answer in Joules or in Watt-hours.)

(Try not to use the internet. If you have earnestly tried for 5 minutes and feel really stuck, you may use the internet to look up one relevant number that you feel like a different reasonable version of you would know off the top of their head, but you should still write down a first guess before doing that.)

If the above happens to be easy for you, e.g. because you basically know the answer, do “how many bits of surprisal does English have per word if you only see the previous word when predicting?” instead.

My chain of thought (+ speculation) on the problem

- Hmm, do I know anything about the energy use of anything relevant/similar?
 - Maybe this thought is in part coming from this sort of thought having been useful on Fermi estimation problems? Just some RL thing? What is the unit that gets boosted? It'd be interesting to know in more detail how that works.
 - But the thought also seems pretty obvious without having done fermi estimation before — it could just be pulled in by some sort of non-fermi-specific “ask a similar question” circuit, say. Maybe it'd be interesting to look at how my thinking on fermi problems changes with more practice to understand the impact of fermi-specific learning better.
- I'm sure I've seen the numbers for the power use of new data centers, hmm but can't seem to remember...
- There's also some fact about how in 5 years or something the power use of AI would be like 25% of all US power use? One could get an answer from this fact if one knew the doubling time and assumed an about even split between inference and training power use (assuming steady growth which might be fine). Is there any way I could figure out the doubling time? I could just use some number like 1/1000 for the current thing... This feels like too many guesses.
 - There's some sort of thing where I'm trying to construct a bridge from my knowledge to the variable sought. There's some search for relevant pieces of knowledge and a search for how to bridge the gap from them. I think the relevant pieces of knowledge come from sth like running a search with a template of “AI power use is x”? This prompts: what sort of memory data structure would support running that search quickly? It's really cool that we can do this semantic search so well. Maybe also think about how we think when guessing in 20 questions. I remember Minsky has some ideas on this.
- What relevant stuff do I even know about GPT-4? I looked up earlier that it was released in March 2023, so maybe trained around the end of 2022? This could help if I had a power use law... I might know its training FLOPs? I think people want to put international AI ban treaty total training FLOP thresholds around 10^{22} or 10^{24} or something. IIRC current models are probably already above at least the lower threshold and maybe the higher one. Maybe I'll say 10^{22} for GPT-4 training.
 - It looks like now I'm doing something similar but starting from a search with a template for GPT-4. I wonder if there's something interesting to be said about what structures make me have this skill / how one gains this skill?

My chain of thought on the problem (pt 2)

- I'm feeling kinda good about the FLOP approach because maybe I can guess Joules per FLOP (seems plausible this is a nice number). I know my laptop is order 50 watts. How many FLOP per sec is my laptop?
 - The question FLOP per sec is just required
- I think that in programming contests a rule of thumb is something like 10^6 or 10^7 serial operations per second. I think a laptop actually does many more FLOPs per sec because that's more low-level and also there's many threads. I feel like saying 10^{10} ? This would give that I need 10^{12} laptop-seconds to make GPT-4, which is $50 \times 10^{12} = 5 \times 10^{13}$ J
- I should probably try another way to estimate. I think maybe GPT-4 training was run on the equivalent of like 100k computers for 3 months? This would give $100 \times 100000 \times 3 \times 30 \times 24 \times 60 \times 60 = 8 \times 10^{13}$ J.

Ok let's look up the answer. Some guy online says it's like 2×10^{14} J = 6×10^{10} W h. Ok oops I was really underestimating the FLOP use of the training run but I guess luckily this was canceled out by an underestimate of the FLOPs of my laptop lol + maybe also higher efficiency of data centers? I guess the second estimate wasn't really hurt by this.

A thought when looking at the transcript after

- Maybe one can think of the process quite well as having a model of the situation, having a goal variable and “known” variables, and somehow being able to make clever picks of variables to consider to make progress from known variables to goal variables?
- Sometimes it feels very easy to introduce a new variable, eg when you have known x and goal y then obviously x/y is a useful variable to introduce if you can guess its value. One could hypothesize that there is just some trained NN-like thing suggesting variables? Maybe this working depends on having good representations of the situation and of variables. What sorts of representations would we want to have? Maybe one should study some toy ideal setting of this.

Math problem inspection exercise

- Find a math/physics/CS problem which you expect would probably take you 5–25 minutes to solve.
 - Pick one that would require some “creativity”/“ingenuity” to solve. Like, pick one that you wouldn’t feel like you already just immediately know a “solution method” for.
 - It’s fine if you don’t end up solving your problem.
- Schelling choices by difficulty tier (probably most people should pick 1–3):
 - Tier 1 (recreational fun math?): One morning a monk sets out at sunrise to climb a path up the mountain to reach the temple at the summit. He arrives at the temple just before sunset. A few days later, he leaves the temple at sunrise to descend the mountain, traveling somewhat faster since it is downhill. Show that there is a spot along the path that the monk will occupy at precisely the same time of day on both trips.
 - Tier 2 (middle school olympiad level?): There are 6 people in a room. Each pair is either on good terms or on bad terms (and the feeling is always mutual). Prove (ie argue compellingly) that there has to be a triple where each pair is on good terms or a triple where each pair is on bad terms.
 - Tier 3 (high school national olympiad level?): There are people labeled $1, 2, 3, \dots$, and each pair is either on good terms or on bad terms. Show that there is an infinite set of people on good terms or an infinite set of people on bad terms.
 - Tier 4 (maybe IMO p1 level but less work?): There are n points on the plane. Each triple makes a triangle of area < 1 . Show that all the points fit in a triangle of area < 4 .
 - Tier 5: Pick some appropriate problem from the 2025 Putnam/IMO/EuPhO.
- Run your inspection routine on solving your problem.

Some more sources on inspecting thinking on problems

- [A prompt I wrote for myself for inspect-solving physics olympiad problems.](#)
- [A pretty detailed inspected transcript from Mariven.](#)
- George Pólya (a Stanford math professor) spent years doing his version of this sort of thing. He wrote multiple books on it:
 - A popular book with problem-solving advice: [How to Solve It](#)
 - A more academic+detailed+interesting version: Mathematics and Plausible Reasoning
- Alan Schoenfeld (a mathematician turned Berkeley math education professor) built on Pólya's work, doing research on how to teach creative mathematical problem-solving. See his book Mathematical Problem Solving.

Studying some mental phenomena

One other version of the proposed investigation is taking some potential “mental phenomenon/process” and speculating about its nature:

- How might one parse a visual scene into objects? See the book *Pattern Theory: The Stochastic Analysis of Real-World Signals* by Mumford for some ideas.
- You often have a mental model of a situation, that lets you easily see/infer some stuff. What sort of thing is that? Sometimes, adding a new “auxiliary question to consider” expands your model to suddenly easily let you answer some question of interest (for example, when someone gives you the hint “consider the outcircle of the triangle ABC” in a geometry problem). How does that work? (See [AlphaGeometry2](#) for one answer.)
- As you read the first paragraph of a novel, some sort of mental picture gets constructed. What sort of thing is that? What does it let you do? Maybe it gets assembled from the meanings of sentences, which in turn get assembled from the meanings of words? What sorts of things might these be? And how might the meanings of words be learned?
- What sort of mental thing is an analogy? See the book *Surfaces and Essences: Analogy as the Fuel and Fire of Thinking* by Douglas Hofstadter and Emmanuel Sander for some ideas.

Studying some mental phenomena (pt 2)

- You can gain the informal notion of a valid mathematical proof from a teacher who would occasionally confusedly tell you that an invalid argument is a valid mathematical proof. You can gain the informal notion of an integral from a teacher who sometimes makes mistakes when integrating. How might that work? What sort of notion-gaining algorithm could let one not repeat the mistakes of one's teacher?
- What sort of thing is a human concept? What is its “type signature”?
- What is it for some concepts to be good? Can you write down some sort of objective function capturing the goodness of a concept or system of concepts?
- Take some concepts from mathematics or physics and investigate where they come from, either historically or just thinking through how one might invent them. See the paper The Concept of Function up to the Middle of the 19th Century by Youschkevitch, the book The Emergence of Probability by Ian Hacking, and the book Inventing Temperature by Hasok Chang.

Some more versions of the proposed investigation

- Suppose you're doing long-term research on some topic or question. Write down your understanding of it again every once in a while. Examine these later to analyze how your understanding improves/changes over days/weeks/months/years. What is it for your understanding to have improved? What things that you did were most responsible for your understanding improving?
- Think through some examples of the historical process of mathematical or scientific progress. See Imre Lakatos's [Proofs and Refutations](#) which analyzes the process of mathematical discovery via a case study of the history of work on Euler's polyhedral formula.
- How do your values guide your actions? How do values guide action in society broadly? What sort of thing are these "values"? For a more concrete example: it'd be interesting to know [if an AI trained to do some "abstract" thing like proving theorems or predicting serial data would care about real-world stuff](#). I claim that if you were in a simulation, you would want to do some big stuff outside that simulation; I further claim that it would be useful to understand why, to answer this AI question.
- Write down [some vignettes capturing important aspects of how "value development" works in you](#).
- You can try to run the proposed inspection protocol basically whenever you're solving a task.

Some further readings

For an example of what this sort of investigation can lead to in the context of AI alignment, see various of [Tsvi Benson-Tilsen's Alignment Forum posts](#) from 2023.

Thank you for listening!

And thanks to Simon Skade for teaching me this problem-solving inspection thing.